Variational Inference, Variational Autoencoders, and Langevin Dynamics

Slide Set 5 — ML & Deep Learning Course

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Outline

1. Learning Distributions
2. Entropy and KL Divergence
3. The ELBO and Variational Inference
4. Variational Autoencoder
5. Langevin Dynamics

From Predictors to Distributions

Classical ML (so far)

Learn a function $f : \mathcal{X} \to \mathcal{Y}$ minimizing expected loss:

$$\min_f \mathbb{E}[(y - f(x))^2]$$

Gives a **point estimate**, no uncertainty, no data generation.

Generative / Probabilistic ML

Learn the full **joint distribution** $p(\mathbf{x})$ or $p(\mathbf{x}, \mathbf{z})$:

- **Density estimation**: what is $p(\mathbf{x})$?
- **Generation**: sample new $\mathbf{x} \sim p(\mathbf{x})$
- **Causal inference**: model interventions $p(\mathbf{x}|\mathrm{do}(z = z_0))$

Latent Variable Models

Posit hidden variables $\mathbf{z}$ governing the data:

$$\mathbf{z} \sim p(\mathbf{z}) \quad \text{(prior)}$$
$$\mathbf{x}|\mathbf{z} \sim p_\theta(\mathbf{x}|\mathbf{z}) \quad \text{(likelihood)}$$

Marginal:

$$p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{x}|\mathbf{z})\, p(\mathbf{z})\, d\mathbf{z}$$

The core difficulty

The integral $\int p(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$ is **intractable** for complex p_θ.

The posterior $p(\mathbf{z}|\mathbf{x}) = \frac{p(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p(\mathbf{x})}$ also.

The Inference Problem: Two Algorithmic Approaches

Why is inference hard?

Computing the posterior $p(\mathbf{z}|\mathbf{x})$ requires the normalising constant:

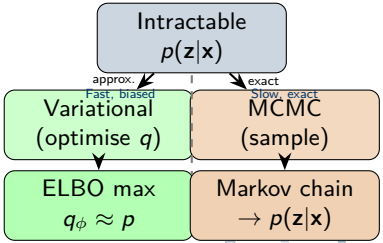
$$p(\mathbf{z}|\mathbf{x}) = \frac{p(\mathbf{x}|\mathbf{z})\, p(\mathbf{z})}{p(\mathbf{x})}, \quad p(\mathbf{x}) = \int p(\mathbf{x}|\mathbf{z})\, p(\mathbf{z})\, d\mathbf{z}$$

Markov Chain Monte Carlo

- Construct a Markov chain with stationary distribution $p(\mathbf{z}|\mathbf{x})$
- Run the chain; collect samples.
- **Pros:** Asymptotically exact;
- **Cons:** Slow; hard to diagnose convergence;

Variational Methods

- **Recast inference as optimisation**
- Choose a tractable family $\mathcal{Q}$ of distributions
- Find $q^* \in \mathcal{Q}$ closest to $p(\mathbf{z}|\mathbf{x})$.
- **Pros:** Fast; GPU-friendly; scales to large data
- **Cons:** Biased non-convex.



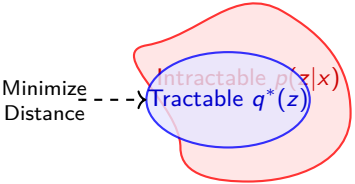
What is Variational Inference? (The Big Picture)

The Core Idea: Inference as Optimization

In Bayesian statistics, we often need to calculate a posterior distribution $p(z|x)$. For complex models, this requires computing an integral that is mathematically impossible (intractable).
Variational Inference (VI) solves this by turning the *integration* problem into an *optimization* problem.

The 3-Step Recipe

1. **The Target:** We have an intractable, highly complex posterior $p(z|x)$.
2. **The Family:** We propose a family $\mathcal{Q}$ of simple, well-behaved distributions (like Gaussians).
3. **The Optimization:** We search through $\mathcal{Q}$ to find the specific distribution $q^*(z)$ that is mathematically "closest" to the true posterior.



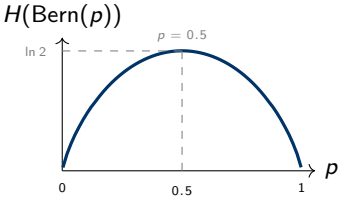
Shannon Entropy

Entropy of a discrete distribution

For a distribution p over outcomes $\mathcal{X}$:

$$H(p) = -\sum_{x \in \mathcal{X}} p(x) \log p(x) = \mathbb{E}_p[-\log p(X)]$$

Measures the *average surprise* (uncertainty) in p.



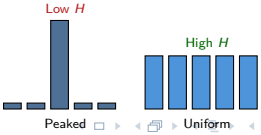
Continuous: Differential Entropy

$$h(p) = -\int p(x) \log p(x)\, dx$$

Example: $X \sim \mathcal{N}(\mu, \sigma^2)$:

$$h(\mathcal{N}) = \frac{1}{2} \log(2\pi e \sigma^2)$$

Interpretation: $H(\text{Bern}(p))$ is highest when the coin is fair. maximum uncertainty. Zero at $p = 0$ or $p = 1$ (certain outcome).



KL Divergence: Definition and Properties

Kullback–Leibler Divergence

$$\mathrm{KL}(p\|q) = \mathbb{E}_p\left[\log \frac{p(X)}{q(X)}\right] = \sum_x p(x) \log \frac{p(x)}{q(x)}$$

Measures how much information is lost when q is used to approximate p.

Theorem 1 (Gibbs' Inequality)

$\mathrm{KL}(p\|q) \geq 0$, *with equality iff* $p = q$ *almost everywhere.*

Relation to entropy

$$\mathrm{KL}(p\|q) = \underbrace{-H(p)}_{\text{neg. entropy}} + \underbrace{H(p,q)}_{\text{cross-entropy}}$$

where $H(p,q) = -\mathbb{E}_p[\log q(X)]$.

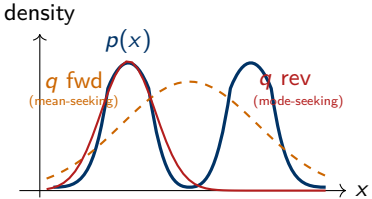
KL is *not* a distance

1. **Asymmetric**: $\mathrm{KL}(p\|q) \neq \mathrm{KL}(q\|p)$ in general
2. **No triangle inequality**: $\exists\, p, q, r$ with $\mathrm{KL}(p\|r) > \mathrm{KL}(p\|q) + \mathrm{KL}(q\|r)$

Note: the *Jensen–Shannon* divergence $\mathrm{JSD}(p\|q) = \frac{1}{2}\mathrm{KL}(p\|m) + \frac{1}{2}\mathrm{KL}(q\|m)$, $m = \frac{p+q}{2}$, *is* a metric.

Forward vs Reverse KL: The Asymmetry Matters

Setting: True distribution p is bimodal.
Approximate with a unimodal q.



Forward KL: $\min_q \mathrm{KL}(p\|q)$

$$\mathrm{KL}(p\|q) = \mathbb{E}_p\left[\log \frac{p(x)}{q(x)}\right]$$

The penalty is large wherever $p(x) > 0$ but $q(x) \approx 0$. q spreads out to cover the support of p.

Reverse KL: $\min_q \mathrm{KL}(q\|p)$

$$\mathrm{KL}(q\|p) = \mathbb{E}_q\left[\log \frac{q(x)}{p(x)}\right]$$

The penalty is large where $q(x) > 0$ but $p(x) \approx 0$.

q **avoids placing mass where** p **is small**.

Consequence for VI

Variational Inference minimizes $\mathrm{KL}(q\|p)$ (reverse KL) for tractability. This causes **mode collapse**: q represents only one mode.

Maximum Likelihood and the Evidence

MLE objective

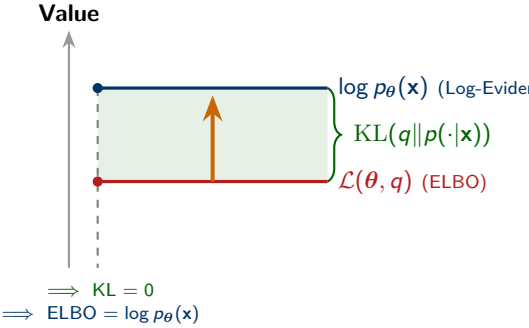
Given i.i.d. data $\mathbf{x}_1, \ldots, \mathbf{x}_n$, maximize:

$$\max_{\theta} \frac{1}{n} \sum_{i=1}^{n} \log p_\theta(\mathbf{x}_i)$$

$\log p_\theta(\mathbf{x}) = \log \int p_\theta(\mathbf{x}|\mathbf{z}) p(\mathbf{z})\, d\mathbf{z}$ is the **log-evidence** (or log-marginal likelihood).

Key decomposition: for any distribution $q(\mathbf{z})$,

$$\log p_\theta(\mathbf{x}) = \underbrace{\mathcal{L}(\theta, q)}_{\text{ELBO}} + \underbrace{\mathrm{KL}(q(\mathbf{z}) \| p_\theta(\mathbf{z}|\mathbf{x}))}_{\geq 0}$$



Interpretation

The ELBO is tight ($= \log p_\theta(\mathbf{x})$) iff $q(\mathbf{z}) = p_\theta(\mathbf{z}|\mathbf{x})$.

ELBO: Full Derivation via Jensen's Inequality

Jensen's inequality derivation

$$
\begin{aligned}
\log p_\theta(\mathbf{x}) &= \log \int p_\theta(\mathbf{x}, \mathbf{z})\, d\mathbf{z} \\
&= \log \int \frac{p_\theta(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} q_\phi(\mathbf{z}|\mathbf{x})\, d\mathbf{z} \\
&\geq \int q_\phi(\mathbf{z}|\mathbf{x}) \log \frac{p_\theta(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\, d\mathbf{z} \quad \text{(Jensen)} \\
&= \mathbb{E}_{q_\phi}\left[\log \frac{p_\theta(\mathbf{x}|\mathbf{z})\, p(\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\right] \\
&= \underbrace{\mathbb{E}_{q_\phi}[\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{reconstruction term}} - \underbrace{\mathrm{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))}_{\text{regularization term}}
\end{aligned}
$$

The Gibbs Variational Principle

Energy-based models

Many probabilistic models take the form:

$$p(\mathbf{x}, \mathbf{z}) = \frac{1}{Z} \exp(-E(\mathbf{x}, \mathbf{z})), \quad Z = \int e^{-E} \, d\mathbf{x} \, d\mathbf{z}$$

E: energy function; Z: intractable partition function. The posterior $p(\mathbf{z}|\mathbf{x})$ is equally intractable.

Theorem 2 (Gibbs Variational Principle)

For any energy E and any distribution $q(\mathbf{z})$:

$$\log Z = \max_q \left[H(q) + \mathbb{E}_q[-E(\mathbf{x}, \mathbf{z})] \right]$$

The maximum is attained at $q = p(\mathbf{z}|\mathbf{x})$.

Connection to the ELBO

Identifying $-E(\mathbf{x}, \mathbf{z}) = \log p(\mathbf{x}, \mathbf{z})$:

$$\begin{aligned} \mathcal{L}(q) &= H(q) + \mathbb{E}_q[\log p(\mathbf{x}, \mathbf{z})] \\ &= \mathbb{E}_q[\log p(\mathbf{x}, \mathbf{z})] - \mathbb{E}_q[\log q(\mathbf{z})] \\ &= \mathbb{E}_q\left[\log \frac{p(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})}\right] \end{aligned}$$

This is precisely the **ELBO**.

Two views of the ELBO

- **Probabilistic**: lower bound on log-evidence.
- **Physics**: negative variational free energy; maximizing ELBO = finding the distribution that balances high entropy with low energy.

Variational Autoencoder: Setup

VAE Generative Model

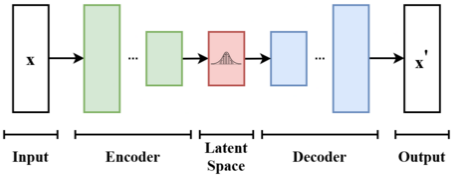
Prior: $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$

Encoder (approximate posterior):

$$q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\phi(\mathbf{x}), \operatorname{diag}(\boldsymbol{\sigma}^2_\phi(\mathbf{x})))$$

Decoder (likelihood):

$$p_\theta(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\boldsymbol{\mu}_\theta(\mathbf{z}), \sigma^2\mathbf{I})$$



ELBO for VAE

$$\mathcal{L}(\boldsymbol{\theta}, \phi; \mathbf{x}) = \underbrace{\mathbb{E}_{q_\phi}[\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{reconstruction}} - \underbrace{\mathrm{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \| \mathcal{N}(\mathbf{0}, \mathbf{I}))}_{\text{regularization}}$$

The KL term has a **closed form** for diagonal Gaussians!

The Reparameterization Trick

The Problem

The ELBO contains $\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\cdot]$. To compute gradients w.r.t. ϕ, we must differentiate *through* the sampling operation $\mathbf{z} \sim q_\phi$. This is **not directly possible**, sampling is a discontinuous, non-differentiable operation.

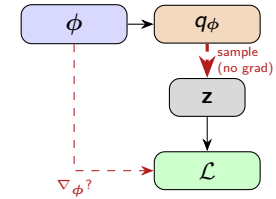
Reparameterization (Kingma & Welling 2013)

Write $\mathbf{z}$ as a **deterministic** function of ϕ and a noise variable $\boldsymbol{\xi}$:

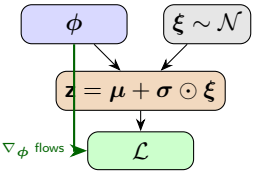
$$\mathbf{z} = \boldsymbol{\mu}_\phi(\mathbf{x}) + \boldsymbol{\sigma}_\phi(\mathbf{x}) \odot \boldsymbol{\xi}, \quad \boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

Now $\nabla_\phi \mathbf{z}$ is well-defined and the gradient flows through $\boldsymbol{\mu}_\phi$ and $\boldsymbol{\sigma}_\phi$.

Without reparam.



With reparam.



VAE: Closed-Form KL and Full Loss

Closed-form KL for diagonal Gaussians

With $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \mathrm{diag}(\boldsymbol{\sigma}^2))$ and $p(\mathbf{z}) = \mathcal{N}(0, I)$:

$$\mathrm{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})) = \frac{1}{2} \sum_{j=1}^{k} \left(\mu_j^2 + \sigma_j^2 - 1 - \log \sigma_j^2 \right)$$

Using $\log \sigma_j^2$ (log-variance) in the network for numerical stability.

Full VAE Training Loss (per sample)

$$\mathcal{L}_{\mathrm{VAE}} = \underbrace{-\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{reconstruction loss}} + \underbrace{\mathrm{KL}(q_\phi \| p)}_{\text{KL regularization}}$$

Algorithm: VAE Training

for each mini-batch $\{\mathbf{x}_i\}_{i=1}^{B}$ **do**
  Compute $(\boldsymbol{\mu}_i, \log \boldsymbol{\sigma}_i^2) \leftarrow \mathrm{Enc}_\phi(\mathbf{x}_i)$
  Sample $\boldsymbol{\xi}_i \sim \mathcal{N}(0, I)$
  $\mathbf{z}_i \leftarrow \boldsymbol{\mu}_i + e^{\log \boldsymbol{\sigma}_i^2 / 2} \odot \boldsymbol{\xi}_i$
  Compute $\hat{\mathbf{x}}_i \leftarrow \mathrm{Dec}_\theta(\mathbf{z}_i)$
  $\mathcal{L}_{\mathrm{recon}} \leftarrow \mathrm{BCE}(\hat{\mathbf{x}}_i, \mathbf{x}_i)$
  $\mathcal{L}_{\mathrm{KL}} \leftarrow \frac{1}{2} \sum_j (\mu_j^2 + \sigma_j^2 - 1 - \log \sigma_j^2)$
  $\nabla_{\theta,\phi}(\mathcal{L}_{\mathrm{recon}} + \mathcal{L}_{\mathrm{KL}})$ via backprop
  Update $\boldsymbol{\theta}, \boldsymbol{\phi}$ with Adam
end for

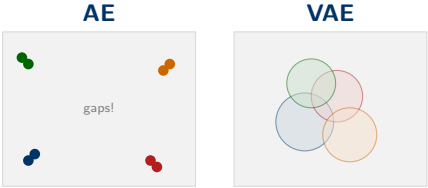
Generation at test time

Sample $\mathbf{z} \sim \mathcal{N}(0, I)$, decode $\hat{\mathbf{x}} = \mu_\theta(\mathbf{z})$.
The structured latent space enables smooth interpolation.

VAE: Geometry and Latent Space

VAE vs standard AE

- **AE**: encodes to a point $\mathbf{z} = e_\phi(\mathbf{x})$. Latent space may have gaps, decoding from a random $\mathbf{z}$ gives nonsense.
- **VAE**: encodes to a distribution $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2 I)$. The KL term pushes encodings toward $\mathcal{N}(0, I)$, the latent space is **dense and continuous** in the manifold support.



Isolated clusters

Dense, interpolable

The Limitations of VAEs (Why look elsewhere?)

- **Blurry Generations:** Restricting the latent space to simple independent Gaussians causes the model to "average out" uncertainty, leading to blurry outputs.
- **Surrogate Objective:** We only optimize a bound (the ELBO), not the exact data likelihood.
- **The Paradigm Shift:** Instead of forcing data through a restrictive, low-dimensional latent bottleneck.

Score Functions: The Key Quantity

Score Function

For a differentiable density $p(\mathbf{x})$, the **score function** is:

$$s(\mathbf{x}) = \nabla_{\mathbf{x}} \log p(\mathbf{x})$$

It is a vector field pointing in the direction of increasing log-probability, toward the modes.

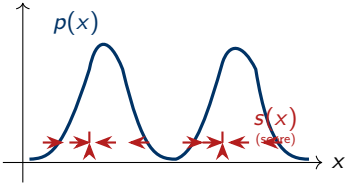
Connection to energy-based models

If $p(\mathbf{x}) = \frac{1}{Z}\exp(-E(\mathbf{x}))$, then:

$$s(\mathbf{x}) = \nabla_{\mathbf{x}} \log p(\mathbf{x}) = -\nabla_{\mathbf{x}} E(\mathbf{x})$$

The score is the **negative energy gradient**, it is defined without knowing Z!.

- We can **learn** $s(\mathbf{x})$ from data without knowing $p(\mathbf{x})$.



Score vectors point toward modes.

In **low-density regions** (between modes), the score is hardest to estimate accurately — a key challenge for diffusion models.

Langevin Dynamics: From Physics to Sampling

Langevin Stochastic Differential Equation

In statistical physics, a particle in a potential $U(\mathbf{x}) = -\log p(\mathbf{x})$ subject to thermal noise obeys:

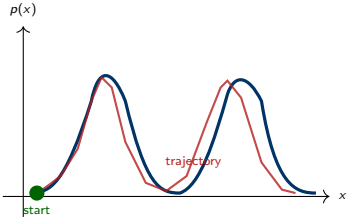
$$d\mathbf{x}_t = -\nabla_{\mathbf{x}} U(\mathbf{x}_t)\, dt + \sqrt{2}\, dW_t$$

$$\Leftrightarrow \quad d\mathbf{x}_t = \nabla_{\mathbf{x}} \log p(\mathbf{x}_t)\, dt + \sqrt{2}\, dW_t$$

The stationary distribution of this SDE is $p(\mathbf{x})$.

Discrete-time (Unadjusted Langevin Algorithm, ULA)

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \frac{\varepsilon}{2} \nabla_{\mathbf{x}} \log p(\mathbf{x}_t) + \sqrt{\varepsilon}\, \boldsymbol{\xi}_t, \quad \boldsymbol{\xi}_t \sim \mathcal{N}(0, I)$$



Two competing forces

1. **Drift**: $\frac{\varepsilon}{2} \nabla_{\mathbf{x}} \log p(\mathbf{x}_t)$ pulls toward modes
2. **Diffusion**: $\sqrt{\varepsilon}\boldsymbol{\xi}_t$ adds random exploration

Balance $\Rightarrow$ chain explores the whole distribution.

Score Matching

Score Matching (Hyvärinen 2005)

We want to estimate $p(\mathbf{x})$ from samples $\{\mathbf{x}_i\}$. Instead of estimating p directly, estimate the **score** $s_\theta(\mathbf{x}) \approx \nabla_\mathbf{x} \log p(\mathbf{x})$. Minimize:

$$J(\boldsymbol{\theta}) = \mathbb{E}_p\left[\|s_\theta(\mathbf{x}) - \nabla_\mathbf{x} \log p(\mathbf{x})\|^2\right]$$

Hyvärinen showed this is equivalent (up to a constant) to:

$$J(\boldsymbol{\theta}) = \mathbb{E}_p\left[\mathrm{tr}(\nabla_\mathbf{x} s_\theta(\mathbf{x})) + \tfrac{1}{2}\|s_\theta(\mathbf{x})\|^2\right]$$

Computable from samples — no $p(\mathbf{x})$ needed.

Denoising Score Matching (Vincent 2011)

Corrupt data: $\tilde{\mathbf{x}} = \mathbf{x} + \boldsymbol{\eta}$, $\boldsymbol{\eta} \sim \mathcal{N}(0, \sigma^2 I)$. Then:

$$\nabla_{\tilde{\mathbf{x}}} \log p_\sigma(\tilde{\mathbf{x}}|\mathbf{x}) = -\frac{\tilde{\mathbf{x}} - \mathbf{x}}{\sigma^2}$$

Train $s_\theta(\tilde{\mathbf{x}})$ to predict this denoising direction:

$$\mathbb{E}_{\mathbf{x},\boldsymbol{\eta}}\left[\left\|s_\theta(\tilde{\mathbf{x}}) + \frac{\boldsymbol{\eta}}{\sigma^2}\right\|^2\right]$$

Avoids expensive Jacobian trace $\mathrm{tr}(\nabla s)$.